

DESIGN AND MANUFACTURING OF A STEERING WHEEL
FOR A SOLAR-POWERED RACING VEHICLE

By

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Design and Manufacturing of a Steering Wheel for a Solar-Powered Racing Vehicle

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Abstract—The steering wheel is the main control mechanism of a car, making reliability, robustness and comfort extremely important when designing this component. The steering wheel size, feel and electronic components are also key to making the driver interface as smooth as possible. As the engineering level increases for formula student competitions, teams want to produce the best car they can to win, and to do this component designs have to be updated to the latest technologies. The main purpose of this paper is to design and manufacture a fully functional steering wheel that will be used in a solar-powered vehicle for different competitions across the US. This paper evaluates the steps taken to design and improve the shape, manufacturing process, and interface of the steering wheel used in Flare, Solar Gators' fourth vehicle. A preliminary analysis was conducted for the top plate which resulted in a maximum stress of 15.05 GPa and a Factor of Safety (FoS) of 4.48. The wheel was subjected to Finite Element Analysis (FEA) Ansys simulations where it was found that the maximum deformation was 2.70×10^{-5} m, the maximum Von Mises stress was 3.67 MPa around the bolt holes, with an FoS of 18.37, and a maximum Von Mises strain of 0.361 also around the bolt holes.

Index Terms — Carbon Fiber, Design, Manufacturing, Prototype, Steering Wheel

I. INTRODUCTION

IN a vehicle, the steering system is the key system used to control the direction of the car's movement and is responsible of ensuring quick and precise response to cornering in the track. The steering wheel serves as the primary interface between the driver and the vehicle, and its design should prioritize ergonomics, structural integrity, and reliability. This paper presents the design, prototyping, manufacturing, and testing procedures followed to enhance the steering wheel of a student run solar-powered racing vehicle design team, Solar Gators.

Formula Sun Grand Prix (FSGP) and American Solar Challenge (ASC) [1] are collegiate competitions where teams design, build, and race solar-powered vehicles. FSGP is a closed-circuit track competition with the objective of completing the most laps withing two days, whereas ASC is a cross-country race across multiple states in the US. Both of these competitions emphasize energy efficiency, strategy and vehicle reliability, making steering wheel performance and driver comfort essential for success. Competitions like these usually have regulations which were considered before starting the design process and can be seen in Fig. 1. The steering wheel

must be one continuous perimeter and although a circular shape is preferred, the upper part above 2/3 and/or lower part below 2/3 of the circumference of the steering wheel may be flat [1]. Additionally, a quick release is needed for fast and safe exiting of the car in under 10 s [1].

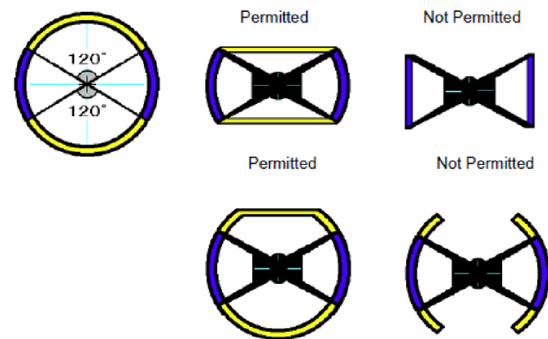


Fig. 1. Steering Wheel FSGP and ASC regulations [1].

Being able to communicate with the team via radio and seeing the speed are crucial to maintaining the safety of the driver. The design process started by evaluating past designs used for the cars. The final design idea was a combination of the past two wheels manufactured, improving both designs by merging the best qualities of each. Two years ago, the steering wheel had a Liquid Cristal Display (LCD) incorporated onto a single Off The Shelf (OTS) carbon fiber ply with two hand supports (Fig. 2). The PCB and screen were covered with a 3D printed enclosure.

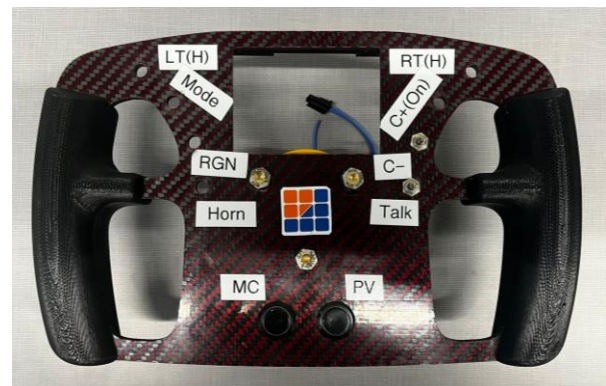


Fig. 2. Previous design, used in 2024. One carbon fiber ply.

The design used in 2025 deviated from the on-wheel LCD, placing it next to the driver external to the steering wheel instead. This design obligated the driver to look away from the road to see the speed at which they were driving among other settings. The steering wheel was made up of two carbon fiber plates manufactured using a mold and performing a wet carbon fiber layup. This design ended in slight delamination after continuous competition use (Fig. 3).



Fig. 3. Previous design, used in 2025. Two carbon fiber layup plates.

The new steering wheel design will incorporate a two-plate structure and an integrated LCD display. The bottom plate will encase the PCB and the screen, eliminating the need of a separate casing. The length of the wheel will be around 28 cm in order to fit the PCB, screen, and quick release used to connect the steering wheel to the steering column. The new design will have 8 buttons with integrated LEDs that light up when pressed. A paddle shifter system will also be designed and used for regenerative braking, converting the kinetic energy from movement into electricity used to charge the car batteries. The hand supports will be improved to make for more comfortable usage. This will be done by reducing the grip diameter by 10%.

User interface was also considered when designing both the physical steering wheel and the electrical components. To improve driver usage, the buttons selected light up when activated and the LCD will display what is currently being used. Additionally, the paddle shifter will allow the driver to actuate and increase the amount of regenerative braking system on the car and reset it to zero braking as well.

Failure modes observed in past designs will be evaluated and analyzed. Based on these, the materials for the new wheel will be chosen to reduce failure chance. Expected failure modes are carbon fiber delamination due to torsional stress and strain around the bolt holes for the quick release.

Finite Element Analysis (FEA) is an essential method in the engineering world to analyze how a component will react under different stress conditions. FEA uses basic Mechanics of Materials (MoM) principles for stress and displacement. A preliminary structural analysis will be performed where two key aspects are considered, stress and deflection. Classical beam theory from MoM can be used to provide accurate estimates compared to the FEA results. The Von Mises principle of stress (1) was used as the basis for stress calculations in the study, which evaluates failure under different loads by providing a single stress value. The Von Mises

principle uses the three principal stresses σ_1 , σ_2 , and σ_3 to obtain the final Von Mises stress, σ_{VM} .

$$\sigma_{VM} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \quad (1)$$

The applied force, F , was estimated by dividing the torque, T , by the steering wheel radius, r (2).

$$F = \frac{T}{r} \quad (2)$$

To estimate the principal normal stress in the z -axis (3), σ_1 , the steering wheel is assumed to act as a rectangular cantilever beam under bending. The load transfer point was assumed to be at the center of the steering wheel, where the quick release is located. It is also assumed that each side of the steering wheel carries half the force due to symmetry. The stress equation can be defined using the bending moment, M , the distance to the neutral axis, y , and the moment of inertia, I . The moment of inertia (4) can be defined using the base, b , and thickness, t of the beam.

$$\sigma_1 = \frac{My}{I} \quad (3)$$

$$I = \frac{bt^3}{12} \quad (4)$$

The Factor of Safety (5), FoS , was calculated by dividing the material strength at break (σ_{break}) by the maximum yield stress (σ_{max}).

$$FoS = \frac{\sigma_{break}}{\sigma_{max}} \quad (5)$$

After performing the preliminary analysis, FEA simulations, using the Ansys software, will be performed on the final design to evaluate the failure modes. The expected results are stress concentrations around the bolt holes for the quick release and around the thinner parts between the hand supports and the body of the steering wheel. Testing for the steering wheel will include prototyping the full model and testing it on the car.

The mechanical assembly of the steering wheel will consist of the two main structure plates, 3D printed spacers for electrical insulation, four hand supports, the quick release mechanism, the paddle shifter, bolts, and nuts.

II. MATERIALS AND METHODS

The design phase began with a review of the previous designs as well as the current racing steering wheels. The previous Solar Gators' steering wheels (Fig. 2-3) differed significantly. The major difference being that one had a screen and the other design didn't. The design requirements for the new design were to implement a screen, accommodation for 8 buttons, and the quick release must be bolted through the center of the steering wheel and allowing the driver to perform egress in less than 10 seconds. The rest of the design decisions were determined by the project engineer.

A. Design Phase

1) Defining Wheel Shape – Front Plate

The steering wheel was designed solely using the SolidWorks Computed-Aided Design (CAD) program. The initial geometry for the first design was based on the front view of the McLaren MCL60 Formula 1 steering wheel (Fig. 4).



Fig. 4. 2023 F1 McLaren steering wheel front view, used as model [2].

The image was imported into SolidWorks and traced using a series of splines and extrusions to obtain a similar shape. Since the regulations for the FSGP require the circumference to be continuous to ensure driver safety, the bottom parts of the steering wheel hand supports were connected together, obtaining a preliminary shape (Fig. 5).

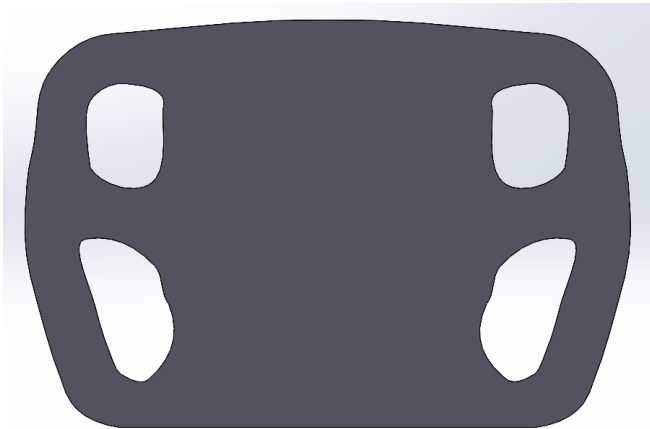


Fig. 5. Preliminary shape, based on F1 steering wheel.

The design process started with the front plate of the wheel. Once the geometry and features were finalized, the dimensions would be taken to define the back plate. Therefore, after having established the preliminary plate shape, fillets and the quick-release bolt holes were added. The prototype was then 3D printed using PETG to determine if the sizing and feel of the design was appropriate (Fig. 6). The size of the previous wheel (Fig. 3) was measured and the dimensions used for this design.

The prototype was evaluated by the car drivers for feedback on ergonomics and usability of the steering wheel. Testing the model inside the car, determined that the overall size of the model was too big to comfortably use inside the cockpit.



Fig. 6. First design prototype.

The raised contour at the hand placement regions was not well received and was redesigned to be a smoother, continuous arc instead. The initial screen position was too low, which resulted in a low quick-release bolts placement. The quick-release bolts were placed at the center of mass so that the center of torque is aligned with the steering column. This ensures proper steering geometry, stability and consistent handling. The screen was then placed above the holes. The updated design with all the changes can be seen below (Fig. 7).

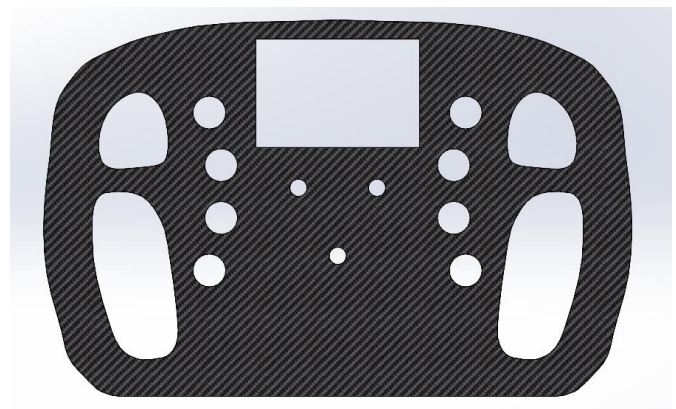


Fig. 7. Second design after feedback changes.

2) Hand Supports

Once the steering wheel geometry was fully defined, the remaining components were designed. The hand supports are essential for driver ergonomics and comfort and are key to enhancing racing performance. The design of these must feel like a mold of the driver's hand while lowering arm fatigue.

The design of the hand supports was more complex than that of the plates, since they required multiple boundary base extrusions and reference planes to achieve the desired shape. Initially, the surface extrusion tool was used; however, it resulted in hollow geometries which weren't suitable for prototyping and caused meshing issues between surfaces. Boundary boss extrusions were used instead, producing a solid body that could be fully meshed.

The front supports (Fig. 8) were designed with a smaller diameter than the rear supports since the thumb arc is smaller than the arc created by the rest of the fingers. Approximate diameters were measured using a flexible measuring tape around the engineer's hand.

The design of one of the handles was made first and then mirrored to obtain the opposite hand. Preliminary bolt and nut holes were added at the top and bottom sections of each hand support. A central ramp was designed with fillets to accommodate the purlicue (area between the thumb and index finger of the hand). The cross section of the supports consisted of semicircular profile with a flat back surface. This allowed for the back side to be bolted flat on top of the steering wheel front plate. The resulting geometry is shown in the figure below (Fig. 8)



Fig. 8. Front hand supports, top view.

The rear hand supports (Fig. 9) were designed following a similar procedure. First designing one handle using boundary extrusions and then mirroring the first design to obtain the opposite hand.

This design featured a less inclined and slightly longer ramp in the middle to accommodate the index finger comfortably. The diameter was bigger than the front design and the shape was more conical than round. Bolt holes were added to the top and bottom sections of each handle.



Fig. 9. Rear hand supports, back view.

3) Paddle Shifters

A regenerative braking system was added to the steering wheel features, and a simple design was developed on SolidWorks [3]. This design incorporated a centrally balanced beam held by a bolt, that would actuate a snap action switch when tilted. The full system assembly (Fig. 10) consisted of a casing with a hole for a magnet, a paddle arm with the opposing magnet pole, and the base which would hold the switch.

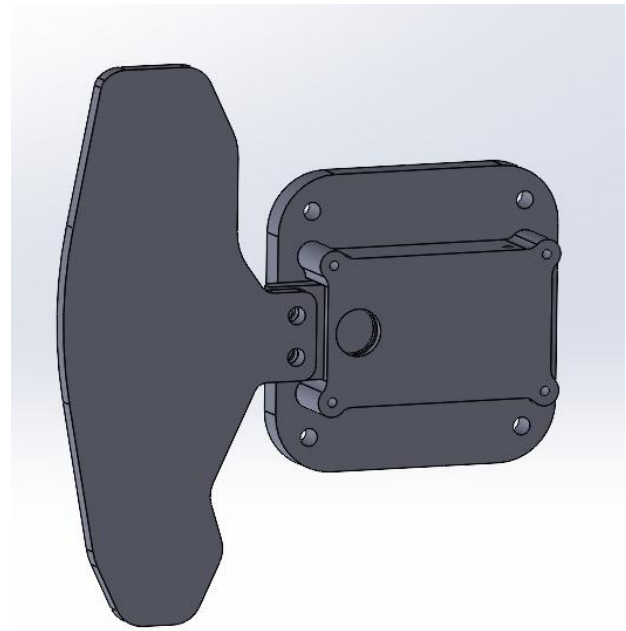


Fig. 10. First paddle shifter design.

However, this design posed several limitations therefore, a new design had to be implemented. Research was conducted on existing racing teams to find possible solutions. A mono arm paddle shifter with one attachment point was designed [4]. The mechanics of the new system would work in exactly the same way as the previous one, a centrally balanced beam held to the casing by a bolt. The casing would now have four magnets slightly to the right of the center bolt and the other two symmetric on the left side. The arm would then contain the opposing poles of the casing magnets, so they attract each other, holding the arm beam in place. The base plate would now contain two snap action switches, one on each end. When the long arm is tilted to one side, it will actuate the corresponding switch. The design can be seen below (Fig. 11).

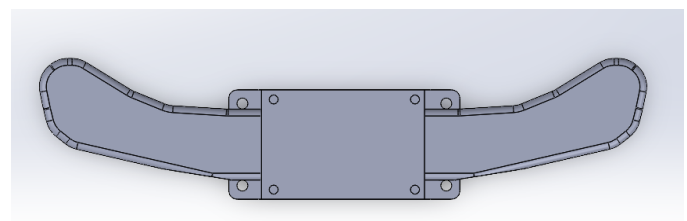


Figure 11. Second and final paddle shifter design, mono arm.

The paddle shifter also contained a spacer block to place the arm at the correct distance to be actuated by the driver. The shifter would be bolted to the steering wheel under the quick release without blocking its actuation.

4) Back Plate Design

The geometry of the front plate was used as the basis for a base extrusion for the handle thickness. Then an extruded cut was performed to create the cavity where the PCB, electronic components, and LCD would be placed. The front view of the back plate clearly shows the cavity where these components will be placed (Fig. 12).

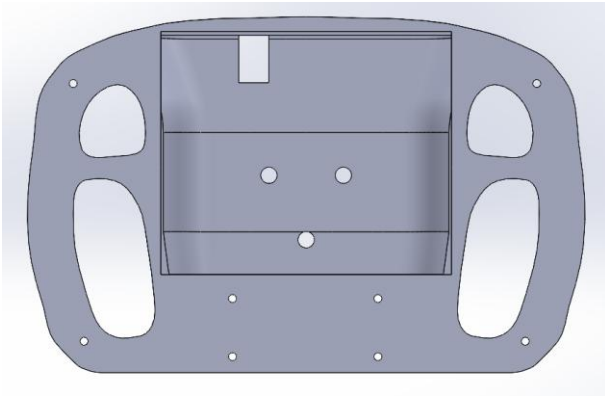


Fig. 12. Back plate, front view.

Then, boundary boss extrusions were applied to create smooth, ramp transitions and a flat casing at the back side of the plate. The casing would also cover the quick release, so bolt holes were added on the surface. The back view (Fig. 13) of the plate shows the geometries explained.

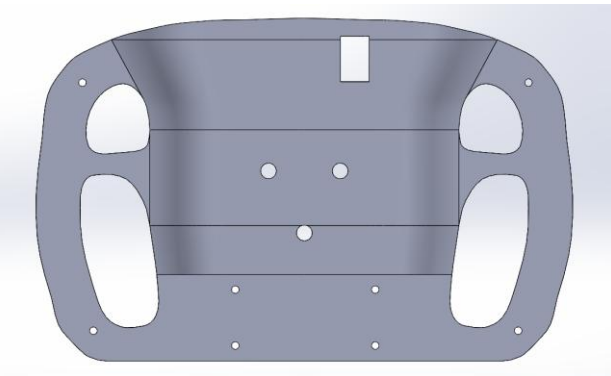


Fig. 13. Back Plate, back view.

Additional bolt holes were added for the hand supports and for the paddle shifter mounting. Holes were also cut for electrical wire exits, two small ones at the end of the boundary extrusion for paddle shifter connection wires and a larger one at the top of the casing extrusion for the LCD screen electrical connection.

5) Final Assembly Model

After finalizing bolt holes and button locations on the front plate, all the parts designed were integrated in an assembly. The final model consists of the front and back plates, front and rear hand supports, the paddle shifter, and the electronic components which include PCB, buttons and LCD (Fig. 14).

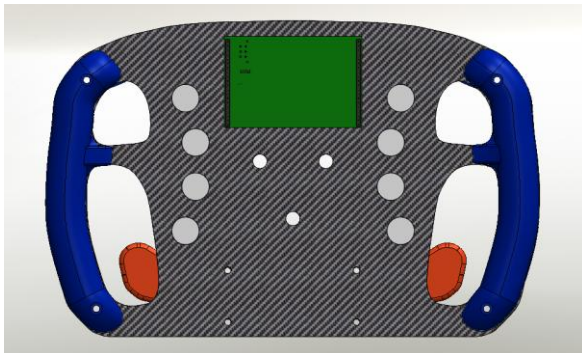


Fig. 14. Final CAD assembly.

Two insulating spacers were also designed. One is placed between the top plate and the top face of the electronic board and the other between the bottom face of the electronic board and the bottom plate. These spacers are required because the plates will be manufactured out of carbon fiber, which is an electric conductor and can cause shorts or damage components on the PCB. The rear isometric view of the final assembly can be seen below (Fig. 15).

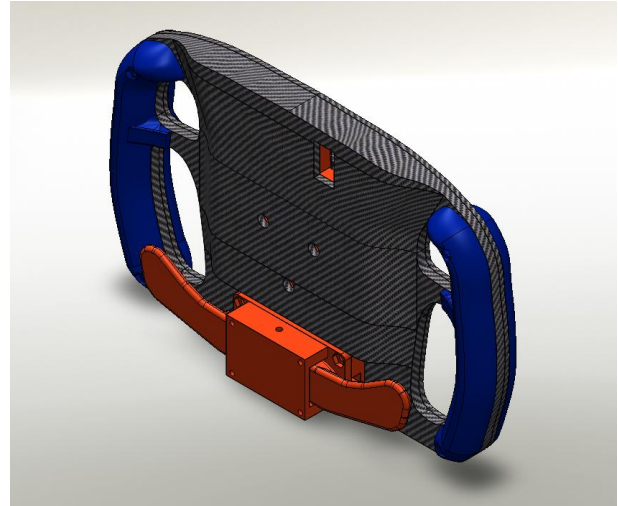


Fig. 15. Back Isometric view.

6) Electric Board Design

The electric board was designed by the electrical engineering team (Fig. 16). The button locations were determined based on the arc movement of the thumb and making sure that all buttons could be reached with ease. The electronic board contains connections for the LCD screen, 8 buttons, and paddle shifter. Actioning the buttons would activate the turn signal indicators, hazard lights, horn, cruise control, and regenerative braking. The screen displays live data essential for the driver during racing and for the team during static charging. Information like the live vehicle speed, turn signal indicators and hazards lights when in use, cruise control status, solar array power levels, main battery and supplemental battery information, and strobe indicator. The board lays on top of the bottom insulating spacer and screwed to it at 4 different locations as seen on Fig. 16.

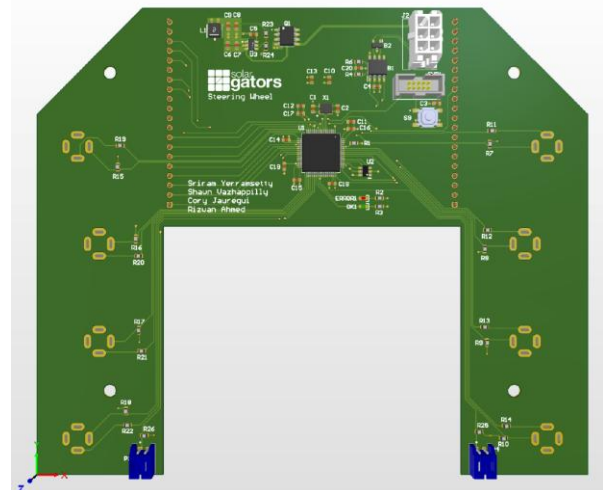


Fig. 16. Steering wheel electric board.

B. Material Selection

PLA was the chosen for the casing and the hand supports since these parts don't have any stress or strain loading. The top plate material was decided to be made of a carbon fiber sandwich wet layup with a Divinycell PVC foam core. A sandwich assembly was chosen because the added core increases stiffness and buckling resistance while keeping the material lightweight. By adding the core, the material acts like an I beam, maximizing structural efficiency and minimizing weight. Carbon Fiber (230GPa), Divinycell H45 145, 2000 epoxy resin and 2100 S slow hardener were the chosen materials for the top plate, with properties shown below in Table I [5].

TABLE I
DIVINYCELL H45 145 MATERIAL PROPERTIES

Quantity	Value
Density	48 kg/m ³
Young's Modulus	55 MPa
Poisson's Ratio	0.4
Bulk Modulus	91.67 MPa
Shear Modulus	19.64 MPa
Tensile Yield Strength	1.40 MPa
Compressive Yield Strength	0.60 MPa

Nylon 12 Carbon Fiber was chosen as the bottom plate material. This material is a polyamide 12 thermoplastic filament reinforced with chopped carbon fiber, 35% by weight. This composite is compatible on a Stratasys FDM 3D printer and has the highest stiffness-to-weight ratio out of all FDM materials. The material properties are shown in Table II [6].

TABLE II
NYLON 12 CF MATERIAL PROPERTIES

Quantity	Value
Density	1190 48 kg/m ³
Young's Modulus X direction	7.91 GPa
Young's Modulus Y direction	7.91 GPa
Young's Modulus Z direction	2.64 GPa
Poisson's Ratio XY	0.33
Poisson's Ratio YZ	0.33
Poisson's Ratio XZ	0.33
Shear Modulus XY	2.97 GPa
Shear Modulus YZ	2.97 GPa
Shear Modulus XZ	2.97 GPa
Tensile Yield Strength X direction	77.5 MPa
Tensile Yield Strength Y direction	77.5 MPa
Tensile Yield Strength Z direction	38.3 MPa
Compressive Yield Strength X direction	-105 MPa
Compressive Yield Strength Y direction	-105 MPa
Compressive Yield Strength Z direction	-135 MPa
Shear XY	152 MPa
Shear YZ	152 MPa
Shear XZ	67.4 MPa

C. FEA Preliminary Analysis

A preliminary analysis was performed before completing the simulations. Due to the geometric complexity of the bottom plate, the simplified preliminary analysis was performed only for the top plate, while both components were assessed using FEA Ansys software. The principal normal stress and the factor of safety were calculated for the Nylon 12 CF part. The

anisotropy introduced by the FDM manufacturing process was not accounted for in the analysis estimation. Additionally, the failure criteria for the carbon fiber composite plate were not evaluated due to orthotropic material behavior.

D. FEA Ansys Simulation Setup

The Ansys Workbench 2025 R2 software was used to evaluate the performance of the design under expected loading conditions. The parts were imported separately into the geometry of an ACP (Ansys Composite PrepPost) block.

The top plate was first sliced along the midplane to create a 2D geometry in preparation for ACP, a simulation module used for modelling and analyzing layered composite structures. The material properties for the top and bottom plates were added into the library (Table I and Table II). Once the materials were selected, meshes were generated with program-controlled element order and an element size of 2×10^{-3} m.

The ACP Pre setup was then configured. First, carbon fiber and Divinycell fabrics were created. An element set and a Rossette were created for the plate. Then an oriented selection set was generated for the new element set. A modelling set was created to define the sandwich panel, two carbon fiber layers, then a foam core, and two carbon layers. Lastly, a solid model was generated to model the part. Once the ACP setup was completed, the ACP model and setup were connected to the model of a static structural block. The static structural block was used to create the necessary FEA simulations.

E. Manufacturing

1) Front Plate Manufacturing

The wheel plates were manufactured differently. The front plate was made of a carbon fiber sandwich layup. The materials used to perform the layup included 2000 epoxy resin, 2100 S slow hardener, isopropyl alcohol 99%, carbon fiber fabric, Divinycell 1/8 in foam sheet cut larger than the steering wheel dimensions, a large polycarbonate plate, measuring cup, brushes, a squeegee, vacuum bag, breather fabric, tacky tape, masking tape, and a pump.

The polycarbonate plate, where the layup would be placed, was cleaned using isopropyl alcohol to prepare the workspace and ensure there was no unwanted debris affecting the part. The surface was then waxed to ease the removal of the layup. Then a 16x20 inch working area was delimited by tacky tape. The tacky tape was then covered with marking tape to avoid getting resin on the tacky tape as shown on Fig. 17.

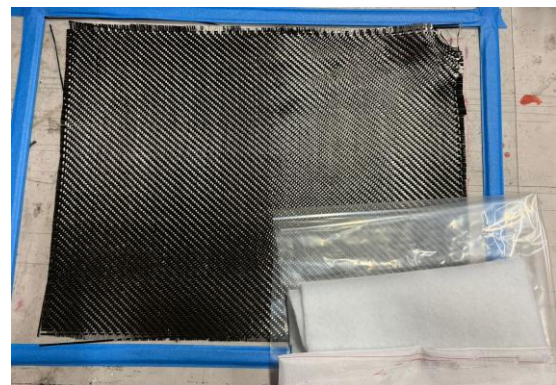


Fig. 17. Carbon fiber sandwich layup materials.

An epoxy-hardener mixture was made by combining 2 parts of epoxy to 1 part of slow hardener. The liquids were mixed until slight bubbling occurred. Once the epoxy mixture was ready, the first carbon fiber lay, a 12x18 inch rectangle was placed on the polycarbonate plate and a generous amount of resin was laid out on the area fabric. Using a squeegee, the resin was spread uniformly until the entire carbon fiber surface was wet. A second ply was laid on top and the fibers were aligned, then more resin was added on top and distributed evenly using a squeegee and making sure the entire surface was covered with an appropriate amount of epoxy.

The next step involved laying the sandwich core, in this case a Divinycell foam sheet, and adding some epoxy mixture on top. This was done to ensure that the next carbon fiber fabric was wet enough to bond. A third lay of carbon fiber fabric was placed on top of the core along with more resin on top of the fabric. The epoxy was spread evenly and a fourth and last carbon fiber fabric was placed on top. A final layer of epoxy was added and evenly distributed at the top.

To prepare the layup for vacuum sealing, a peel ply fabric was placed on top of the carbon fiber and breather fabric was placed on top of the latter. Then the breather fabric was covered with a vacuum bag to seal the layup for curing. The vacuum edges were sealed to the tacky tape making sure there were no air leaks. A vacuum pump was connected to the top of the vacuum bag to evacuate air and consolidate and seal the layup during curing (Fig. 18).

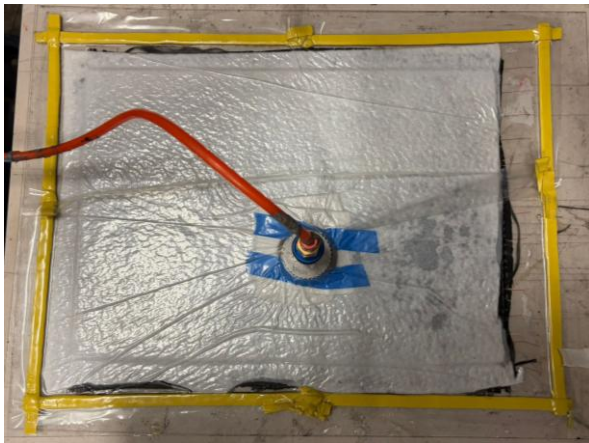


Fig. 18. Carbon Fiber layup vacuum set-up.

III. RESULTS

A. Prototyping Phase

After finalizing the complete steering wheel assembly on SolidWorks, several prototypes were manufactured to validate the designs in terms of dimensions, geometry, and overall ergonomics. The first prototype, (Fig. 6) previously mentioned in the design phase, was critical to identify design issues such as sizing and shape of the steering wheel.

1) Hand Support Prototyping

After completing the hand support designs, a one-third section of the full steering wheel assembly was 3D printed using Matte Polylactic Acid (PLA). This print included both right hand supports, the button placements and both plates

assembled (Fig. 19). This prototype was manufactured to check the feel and shape of the hand supports with the plate thickness added. The feedback obtained from the prototype was positive, however, the drivers requested more rounded edges in the center sections of the hand supports since it could dig into the hands after driving for extended periods of time.



Fig. 19. One-third section of steering wheel, hand support prototype.

2) Paddle Shifter Prototyping

Prototyping of the paddle shifter was required to ensure that the mechanical functionality, and the dimensions were compatible with the steering wheel. At the beginning of the design phase for the paddle shifters, the plan was to use two separate shifter systems (Fig. 10). The first model was 3D printed using PETG to evaluate the system would fit on the wheel and the interaction of the paddle arm with the snap action switch (Fig. 20).



Fig. 20. First paddle shifter design prototype.

While this first prototype worked as intended mechanically, determining the positioning of two shifters without blocking the quick release was impractical. For this reason, the design of the paddle shifter was revised to a mono-arm configuration (Fig. 11). This design could be mounted under the quick release and would only require one mounting location.

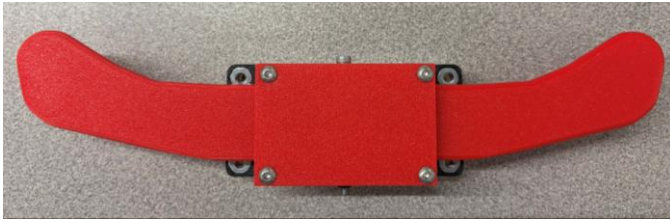


Fig. 21. Mono-arm paddle shifter prototype top view

3) Full Steering Wheel Prototyping

After printing and testing the feel of the hand supports and refining small issues, a full prototype was fabricated before manufacturing the final steering wheel model. Thus, the full assembly prototype was modelled and tested on the car. The front and back plates were printed using a Stratasys Fused Deposition Modelling (FDM) printer and Acrylonitrile Styrene Acrylate (ASA) material. The hand supports were printed using Matte PLA in a Bambu Lab 3D printer.

The full assembly (Fig. 22) included the top hand supports, the top plate, top spacer, electric board, bottom spacer, back plate, rear hand supports, paddle shifter, and the quick release.



Fig. 22. Full steering wheel prototype, front view.

In addition to the plates and the hand supports, the mono arm paddle shifter (Fig. 23) was manufactured in Matte PLA to evaluate its mechanical performance, overall fit, and durability under expected driving conditions. The hand supports, plates, paddle shifter, and quick release were all assembled and secured using bolts.



Fig. 23. Full steering wheel prototype, rear view.

B. Finite Element Analysis Simulations

FEA was carried out on the steering wheel plates to evaluate their structural integrity and their response under an applied torque of 9 Nm. The torque was selected based on previous studies of past simulations carried out on Formula Student vehicles [7].

1) Preliminary Analysis

Prior to performing FEA on the simulation tool, a simplified analytical approximation of the von Mises stress was calculated using Mechanics of Materials principles. Due to the geometric complexity of the bottom plate, the simplified preliminary analysis was performed only for the top plate, while both components were assessed using FEA Ansys software.

The results obtained for the preliminary analysis of the top plate are summarized in Table III. The principal stress for the 9 Nm torque applied resulted in 15.05 MPa at the center of the steering wheel, where the quick release would be positioned.

TABLE III
PRELIMINARY ANALYSIS – TOP PLATE

Symbol	Quantity	Value
T	Torque applied (Nm)	9.00
F	Force applied (N)	62.72
A	Steering Wheel Surface Area (m ²)	0.0775
σ	Principal Normal Stress (MPa)	15.05
M	Bending Moment (N)	9.00
I	Bending Moment of Inertia (m ⁴)	1.495×10^{-9}
y	Distance to Neutral Axis (m)	0.0025
b	Length of the wheel under bending (m)	0.1435
t	Thickness of steering wheel (m)	0.005

2) Ansys FEA Simulation Results

After defining the material properties and ACP geometry in Ansys, a Static Structural analysis block was created. The model and ACP setup from the ACP (Pre) block were connected to the model in the static structure block. The model was opened and the contact between the parts was defined as bonded since the parts would be epoxied. Then the quick release bolt holes were defined as fixed supports. Lastly, a moment was applied about the center of the steering wheel, which coincides with the center of the quick release. The moment was set to 9 Nm about the Z axis. The results included total deformation, equivalent stress, and equivalent elastic strain.

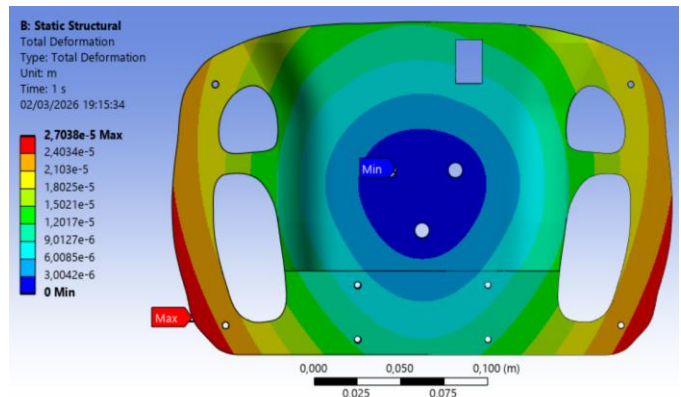


Fig. 24. Back view total deformation simulation due to 9 Nm moment.

The maximum total deformation (Fig. 24) on the steering wheel due to the moment applied was 2.074×10^{-5} m at the edge of the steering wheel hand supports.

The maximum deformation occurs at the lowest edges of the hand supports, with reduced deformation observed throughout the rest of the wheel. The minimal amount of deformation would be centered around the bolt holes, where the quick release would be attached and where the fixed supports were applied in the simulation.

The maximum equivalent (von Mises) stress due to the applied moment was 3.67 MPa, concentrated around the quick release bolt holes. This stress result is lower than the Nylon 12 CF material strength, indicating that the design operates within safe limits. The back view can be seen below (Fig. 25).

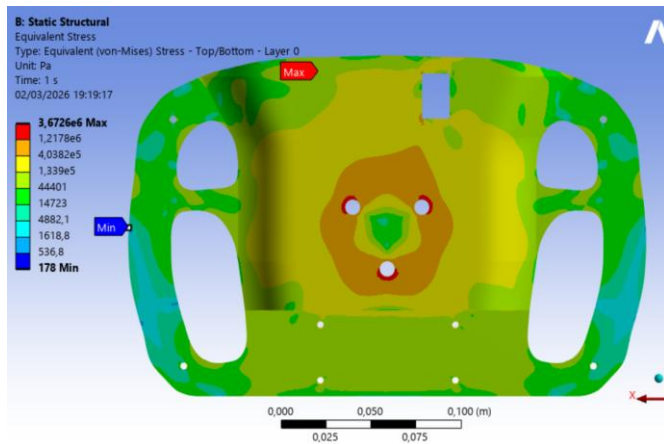


Fig. 25. Back view equivalent stress due to 9 Nm moment.

The maximum equivalent elastic strain due to the applied moment turned out to be 0.361×10^{-3} around the quick release bolt holes, since they are fixed to the steering column. The rest of the steering wheel exhibits negligible strain, indicating a stiff structure. The back view can be seen below (Fig. 26).

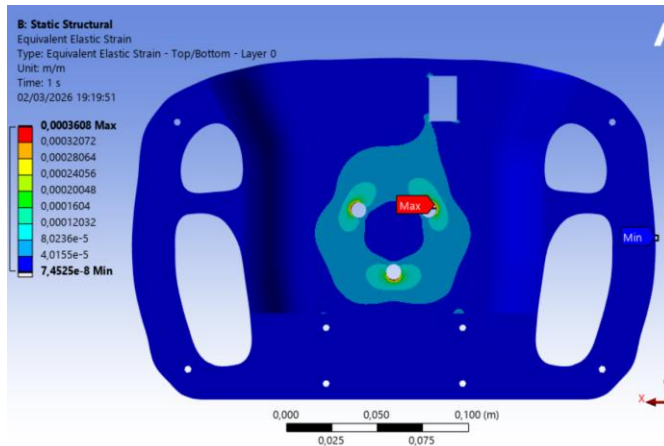


Fig. 26. Back view elastic strain simulation due to 9 Nm moment.

The Factor of Safety obtained for the back plate (5) was calculated using the resultant maximum stress found in the FEA simulation, 3.67 MPa, and the lowest strength at break obtained from the material property sheet [6], 67.4 MPa. This resulted in a Factor of Safety for the back plate of approximately 18.37.

C. Manufacturing Results

1) Front Plate Manufacturing

Following the layup process (Fig. 18), the composite was left to cure for at least 6 hours with the pump continuously operating, compressing the fabrics together and removing air voids. Once the layup was cured, it was then cut into the final steering wheel geometry using a waterjet (Fig. 27) and sanded to remove any sharp edges and surface irregularities.



Fig. 27. Carbon fiber top plate, cut using a waterjet.

2) Back Plate Manufacturing

The back plate was manufactured using a Stratasys Fortus 450mc FDM 3D printer and Nylon 12 Carbon Fiber (CF) filament. The layer height was set to 0.010 in the z direction with solid infill to increase strength. The resultant plate (Fig. 28) exhibits high strength and stiffness and a low weight.

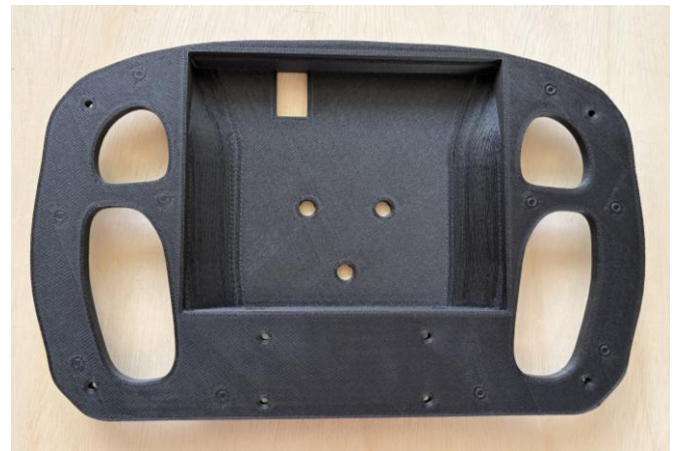


Fig. 28. Back plate 3D printed with Nylon 12 CF.

3) Electrical Spacers Manufacturing

Carbon Fiber is electrically conductive, meaning that when in contact with a working electric circuit, it can conduct current across its surface, causing shorts in the circuit and overheating of components. For this reason, two spacers were 3D printed using Matte PLA.

The spacers would be placed on top and below the electric board. The spacer placed under the PCB separating the bottom plate, was an extrusion of the base plate, with the cavity acting like a mold and with holes to secure the board and for the quick

release and cabling. The top spacer, between the carbon fiber sandwich and the board, was a thin extrusion with holes cut in for the buttons and the quick release bolts, as well as a cavity for the screen placement.

4) Final Steering Wheel Assembly

After manufacturing all components, the steering wheel was assembled. The final assembly consists of the carbon fiber sandwich front plate (Fig. 27), hand supports, two insulating spacers, the Nylon 12 CF back plate, the paddle shifter, quick release mechanism, electric board, LCD display, and bolts. The final product front view can be seen below (Fig. 29).



Fig. 29. Final steering wheel assembly, front view.

The rear view of the assembly can be seen below (Fig. 30).



Fig. 30. Final steering wheel assembly, back view.

D. Testing Phase

The prototype was installed and tested on the car to evaluate fit and driver comfort before manufacturing (Fig. 31). Initial testing revealed a compatibility issue when placing the prototype on the steering column. The column used for the previous steering wheel had an additional extrusion not accounted for in the new design. The interference was resolved by sawing off the extra material. After this modification, the steering wheel fit well and worked as intended both structurally and electrically. Furthermore, driver feedback confirmed that the design was comfortable when driving.



Fig. 31. Steering wheel attached to steering column.

IV. DISCUSSION

The primary objective was to design a steering wheel capable of resisting the steering forces applied when driving and reduce delamination. To do this, a two-plate design was chosen to distribute the loads between the two parts, contrary to the 2024 (Fig. 2) design where all the load was supported by one carbon fiber ply. Additionally, the plate thickness was increased relative to the 2025 design (Fig. 3) and it was decided that the back plate would cover the screen and PCB instead of designing a 3D printed enclosure.

Another key design specification was to enhance the ergonomics and driver comfort during long driving period use. This was addressed through the integration of custom designed hand supports. The design also met the functional requirements, including screen integration, and 8 working buttons. During the design process of the steering wheel, the electrical team requested the integration of a paddle shifter to actuate a new regenerative braking system for the car. This system would convert the kinetic energy from the car into electrical energy to charge the battery pack of the car.

Material selection is of extreme importance as the driver's safety depends on the steering wheel being sufficiently stiff and strong enough to withstand the torque required to control the vehicle. Potential failure modes must therefore be considered when choosing a material. Based on the design of the wheel, most of the stress concentrations are expected to be located at the quick release bolt holes and at the thinnest sections of the hand supports. To reduce these risks, plate thickness was increased relative to previous designs to reduce delamination and fracture.

The torque selection was made based on the results presented in an article published where several Formula Student vehicles were tested. These vehicles required a steering torque between 4 and 8.5 Nm for a diameter larger than 300 mm. Comparing these numbers to the diameter of the steering wheel designed, 283 mm, a maximum torque of 9 Nm was decided to be tested on the FEA simulations.

While the prototype demonstrated satisfactory performance, more testing is recommended to be conducted with the final steering wheel. Future work would include performing a vibration test as well as a shock test to evaluate the real strength of the combined carbon fiber layup and Nylon 12 CF print. By adding strain gauges and using data logging electronics to evaluate driving conditions on the wheel would provide a good insight into how the steering wheel would behave under stress. Additionally, an anisotropic failure criterion for the carbon fiber sandwich panel should be evaluated to more accurately capture the behavior of the steering wheel as a whole. However, based on the FEA simulations conducted and presented above, the steering wheel is expected to be strong and stiff enough to sustain the loads applied during driving and competition.

Other future improvements include replacing the Nylon 12 CF 3D printed back plate with a carbon fiber composite layup. The resultant steering wheel would have an increased strength and would reduce the risk of long-term delamination.

The limitations posed by the first paddle shifter design were that two separate systems were required, one to actuate and increase regenerative braking and another to reset the setting to zero braking. With two models, there was no physical space on the wheel to attach the systems without blocking the quick release actuation. An improvement to the paddle shifter design would be to manufacture the paddle shifter arm out of carbon fiber or Nylon 12 CF. The current PLA model is susceptible to fracture under higher actuation forces; however, carbon fiber would be a stronger and stiffer material capable of withstanding these forces.

V. CONCLUSION

The main objective of this project was to design a stiffer and stronger steering wheel incorporating an LCD display and eight functional buttons. Additional design goals included improving ergonomics, aesthetics, and integrating a paddle shifter for regenerative braking control.

The final design consists of a two- plate steering wheel with integrated hand supports, an LCD display, eight buttons, and a paddle shifter. The top plate was manufactured as a carbon fiber sandwich composite with a Divinycell foam the core, while the back plate was 3D printed using FDM and Nylon 12 carbon fiber filament.

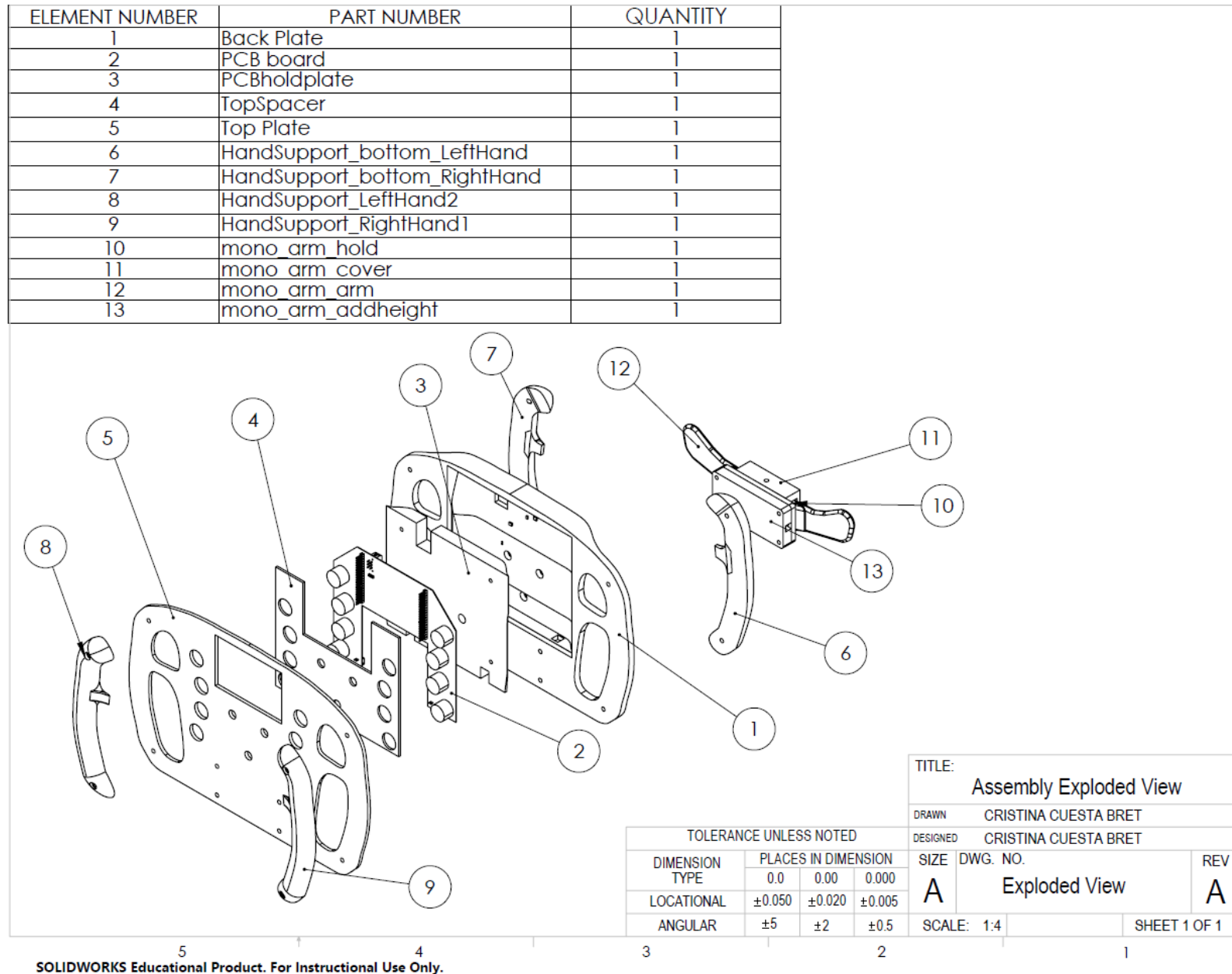
Finite Element Analysis results showed a maximum equivalent stress of 3.67 MPa, a total deformation of 2.074×10^{-5} m, and an equivalent elastic strain of 0.361×10^{-3} . Using the minimum directional strength at break of the Nylon 12 CF, a conservative Factor of Safety for the back plate, of approximately 18 was obtained. These results demonstrate that the steering wheel design is structurally robust and capable of withstanding the applied torque of 9 Nm.

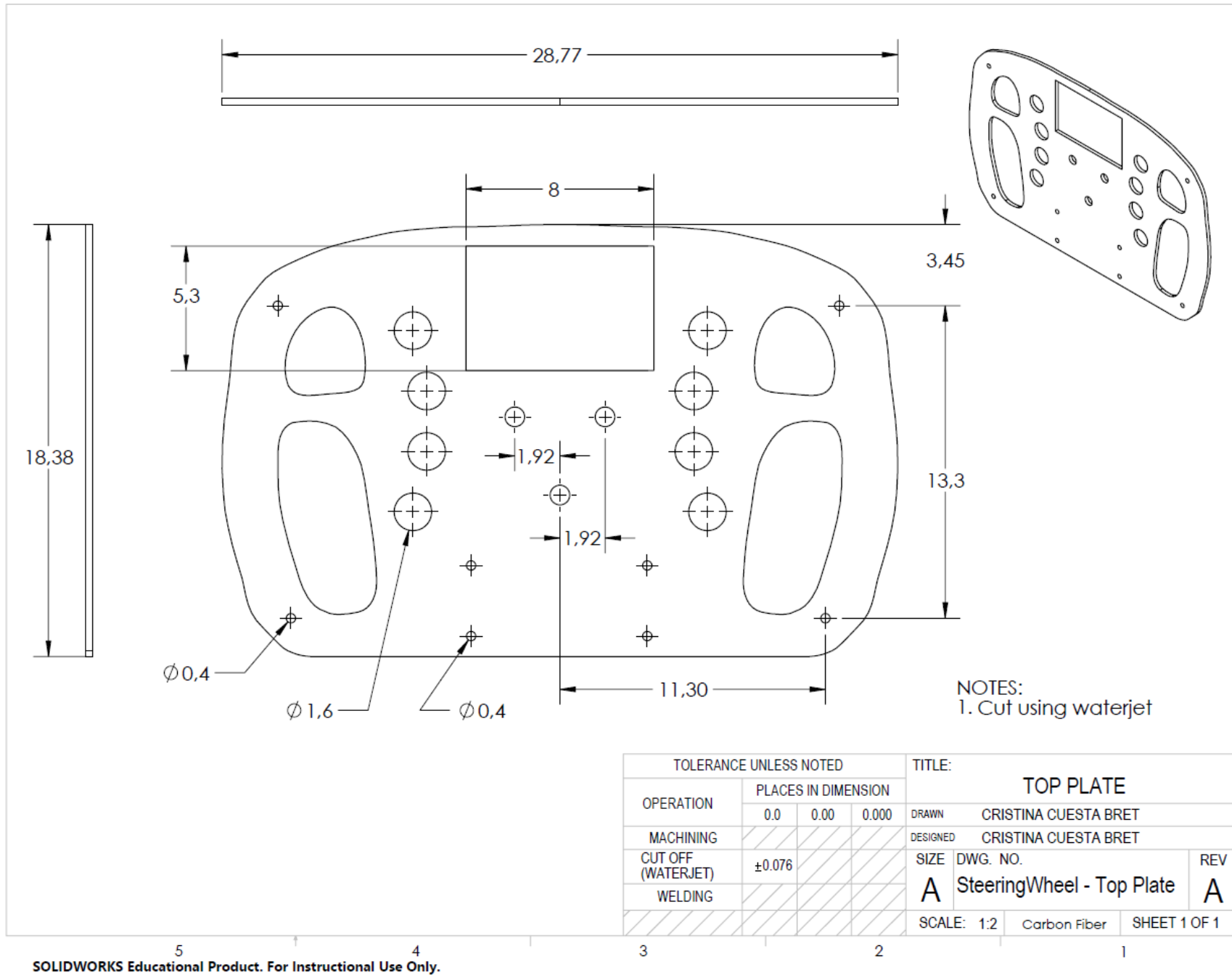
Overall, the proposed design successfully meets the structural, functional, and ergonomic requirements defined for the FSGP and ASC competitions.

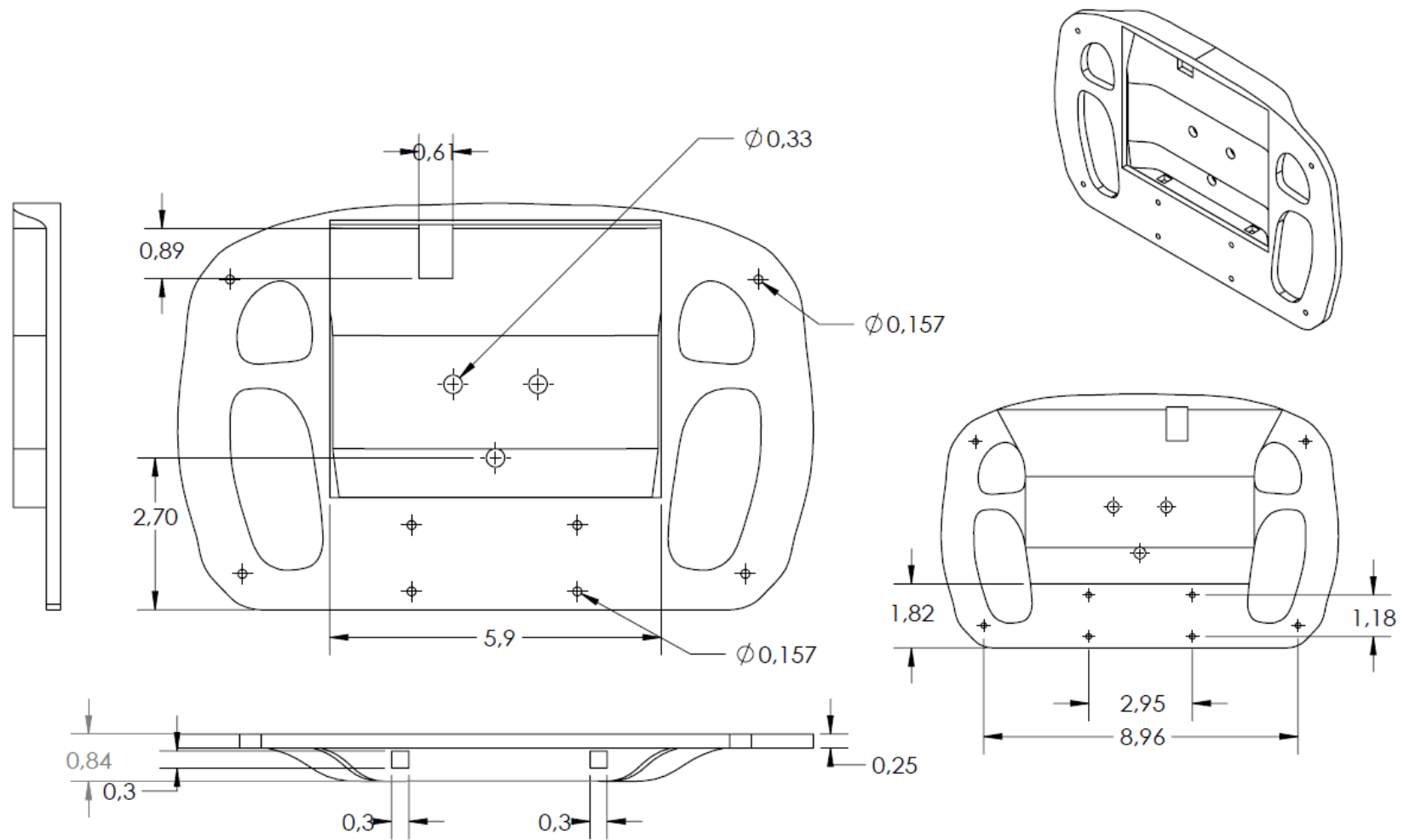
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VI. APPENDIX A. ENGINEERING DRAWINGS



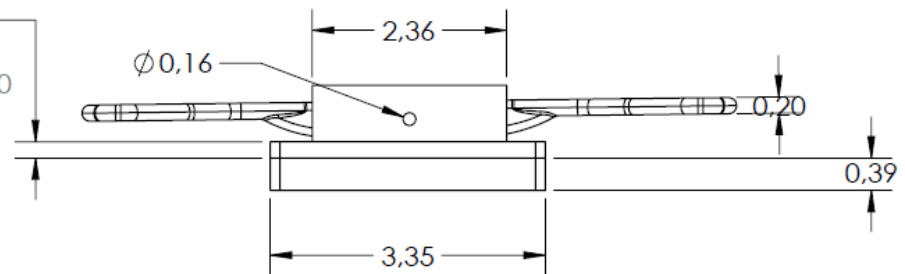
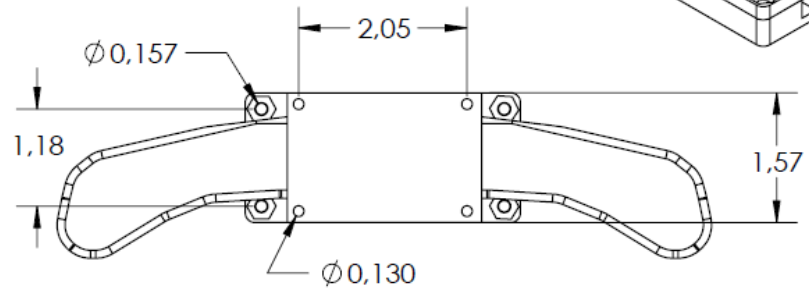
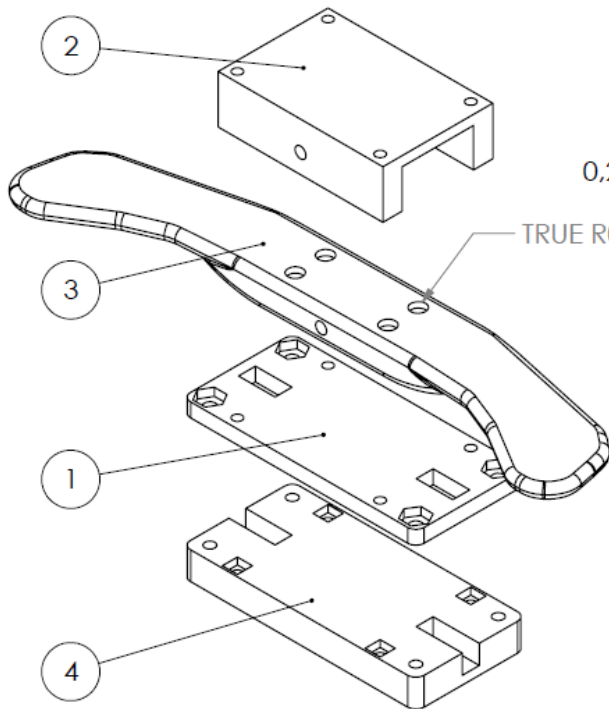




NOTES:
1.3D printed

TOLERANCE UNLESS NOTED				TITLE:		
OPERATION	PLACES IN DIMENSION			Back Plate		
	0.0	0.00	0.000	DRAWN	CRISTINA CUESTA BRET	
MACHINING	± 0.05	± 0.020	± 0.0035	DESIGNED	CRISTINA CUESTA BRET	
				SIZE	DWG. NO.	REV
				A	Steering Wheel - Back Plate	A
				SCALE: 1:2	Nylon 12 CF	SHEET 1 OF 1

ELEMENT NUMBER	PART NUMBER	QUANTITY
1	mono_arm_hold	1
2	mono_arm_cover	1
3	mono_arm_arm	1
4	mono_arm_addheight	1



NOTES:
1. 3D Printed
2. _____

TITLE:				
Paddle Shifter Assembly				
DRAWN		CRISTINA CUESTA BRET		
DESIGNED		CRISTINA CUESTA BRET		
SIZE	DWG. NO.			REV
A	Steering Wheel - Paddle Shifter			A
SCALE: 1:2		SHEET 1 OF 2		